

Summary of Dr. Geraldine L. Cochran's Howard University Physics Colloquium

Title: *Findings from a National Initiative on Transforming Introductory Physics Sequences*

Speaker: Dr. Geraldine L. Cochran, Associate Professor of Physics, The Ohio State University

Dr. Cochran challenged the conventional description of students as “underprepared” for college physics. Her central argument was that differences in preparation frequently reflect unequal educational opportunities rather than differences in intellectual ability. The more constructive question, she proposed, is whether physics departments are adequately prepared to educate the students their institutions have chosen to admit.

Unequal preparation begins before college

Dr. Cochran presented national evidence showing that many students attend high schools that do not offer the full sequence of mathematics and science courses expected by universities. Some schools do not offer calculus, precalculus, or even physics. Nevertheless, colleges may require these subjects before students can enter calculus-based physics.

Consequently, students can be excluded from physics because of opportunities their schools never provided. These inequalities are also correlated with socioeconomic status and race. Dr. Cochran emphasized that when socioeconomic conditions are properly considered, apparent racial differences in mathematical preparation can diminish or disappear. The issue is therefore better understood as an **educational opportunity deficit**, rather than a deficit in students' abilities.

Higher education can either perpetuate these inequalities—by using prior opportunities as gatekeeping requirements—or respond by redesigning instruction.

From “fixing students” to transforming systems

Dr. Cochran distinguished among several responses to students with uneven preparation:

1. Concluding that certain students do not belong in physics or STEM.
2. Requiring remedial courses before allowing students to begin physics.
3. Adding tutoring, bridge programs, or other support around an unchanged course.
4. Redesigning the introductory physics sequence itself.

Tutoring and bridge programs can be valuable, but they still place much of the burden on students. Dr. Cochran advocated the fourth approach: examining whether the curriculum, pace, prerequisites, and teaching practices are appropriate for the students actually enrolled.

This philosophy underlies the NSF-supported **Transforming Introductory Physics Sequences to Support All Students (TIPSSS) Network**. TIPSSS brings together researchers, instructors, graduate students, and advisers to:

- Remove unnecessary barriers to calculus-based physics.
- Integrate mathematical development and quantitative literacy into physics instruction.
- Develop curricular materials that can be adapted to different course structures.
- Study existing institutional models.
- Demonstrate how successful approaches can be implemented elsewhere.

Alternative introductory-physics structures

Participating and related institutions have adopted several models:

- **Foundational or preparatory courses** completed before the conventional sequence.
- **Co-requisite courses** taken alongside introductory physics.
- **Extended sequences** that distribute mechanics over two semesters before electricity and magnetism.
- **Off-ramp or parachute courses**, allowing students who encounter difficulty to transfer into a supportive course without losing an entire semester.

Dr. Cochran favors the extended-sequence model. At Rutgers and Ohio State, additional time allows mechanics, mathematical reasoning, and physical interpretation to be developed more deliberately. Students may also move between extended and traditional pathways as their needs become clearer.

Importantly, an extended sequence should not mean reduced expectations. At Rutgers, students from the extended and traditional pathways subsequently performed comparably when they entered the common electricity-and-magnetism course. Preliminary Ohio State results also suggest that students from the extended sequence are succeeding in the subsequent course, although Dr. Cochran cautioned that the Ohio State sample is still small.

Mathematics is not the same as quantitative literacy

A major theme was the distinction between possessing mathematical skills and being able to use mathematics meaningfully in physics.

Quantitative literacy involves interpreting numbers, equations, signs, units, ratios, and relationships within a physical situation. Experienced physicists routinely ask:

- Does this result make physical sense?
- Are the sign and direction consistent?
- What happens in a limiting case?
- Are the units appropriate?
- How should the mathematical relationship be interpreted physically?

Students do not necessarily acquire these habits in a mathematics course. Some students succeed in calculus but still struggle in physics because the difficulty lies not in performing a calculation, but in connecting the calculation to the physical world.

For example, physicists use negative signs differently in kinematics, work and energy, thermodynamics, and temperature. Instructors may understand these distinctions automatically, while students experience the same symbol appearing to change meaning from one topic to another. Dr. Cochran argued that such reasoning must be taught explicitly rather than treated as something students will simply absorb.

Teaching students to think like physicists

TIPSSS materials begin with accessible situations before introducing formal physics. Students might first reason about the relative numbers or prices of pencils, pennies, and dimes. They learn to translate verbal relationships into equations and to test those equations with simple numerical cases.

The curriculum then transfers those habits into physics. Rather than immediately memorizing a formula for velocity, for example, students may construct a “fastness index” for several cars. This helps them understand why a physical quantity is defined as it is before attaching formal notation to it.

Student development is assessed using the **Physics Inventory of Quantitative Literacy (PIQL)**, which examines reasoning about proportional relationships, covariation, negative signs, and related concepts. Across the participating institutions, courses incorporating quantitative-literacy materials generally produced gains, with statistically significant improvements in the larger samples.

What the preliminary evidence suggests

Dr. Cochran was careful not to make causal claims that the current studies cannot support. Many transformed courses combine several changes: additional instructional time, active learning, undergraduate learning assistants, new curricula, and more explicit mathematical support. The studies have not yet isolated the contribution of each component.

Nevertheless, the emerging findings are encouraging:

- Students improve their quantitative literacy.
- Extended-course students can pass at rates comparable to students in traditional sequences.
- Their subsequent performance suggests that the courses are building preparation rather than merely awarding easier grades.
- Earlier institutional evidence indicates improved persistence and graduation, including for women and students from historically marginalized groups.
- Courses designed around students’ actual preparation and disciplinary interests tend to improve engagement and performance.

TIPSSS is planning longer-term research, including a longitudinal study of Rutgers students, to examine later coursework, engineering progression, time to degree, and graduation.

Central takeaway

Dr. Cochran's message was not that physics programs should lower their standards. It was that departments should teach explicitly the reasoning their standards require.

Students should not be excluded because their previous schools lacked calculus or physics. If universities admit students into STEM programs, those institutions share responsibility for providing credible pathways to success. Transforming introductory physics therefore means maintaining intellectual rigor while reconsidering prerequisites, pacing, course structure, and the often-unspoken reasoning habits that physicists expect students to possess.

In Dr. Cochran's reframing, the question is no longer simply:

“Are these students prepared for physics?”

It is also:

“Are our physics programs prepared to educate these students?”